How much shorter can a valid prediction-powered interval be when labels, unlabeled data and predictor are fixed?

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Abstract

Prediction-powered inference (PPI) and PPI++ give valid confidence intervals from a few labels, many unlabeled records and a fixed predictor. We ask whether, with labels, unlabeled data and predictor all fixed, such an interval for a population mean can be made shorter while keeping 95% coverage, and when. We state the efficiency ceiling (it depends on the predictor only through $E[Y \mid f]$, and PPI++ attains it iff that is affine) and test a cross-fitted calibrated-score estimator and small-sample corrections against the `ppi-python` PPI++ reference with 2000 paired replicates per cell on nine real datasets and 45 synthetic populations. On the real datasets there is essentially no room: the ceiling is within 0.5% of PPI++ for seven datasets, and the calibrated interval is wider than PPI++ in 24 of 26 primary cells (by up to 11% at $n = 50$) because learning the map costs more than it gains. On synthetic populations with clearly nonlinear $E[Y \mid f]$ the calibrated interval is up to 21% shorter, though not always at nominal coverage, and never reaches the ceiling. At small n the PPI++ reference undercovers (0.84 on census income at $n = 50$) and a jackknife-t repair costs 10–15% width; under label–unlabeled shift, unclipped power-tuned weights fail (coverage 0.49 at $n = 400$) while the clipped default survives. The answer is therefore conditional: shortening needs a predictor with nonlinear conditional mean and enough labels to learn it.

1 Introduction

Gold-standard labels are scarce in many sciences, while features and a trained predictor are cheap. Prediction-powered inference (PPI) [Angelopoulos et al., 2023a] and its power-tuned version PPI++ [Angelopoulos et al., 2023b] combine a small labeled sample with predictions on a large unlabeled sample to give confidence intervals that stay valid whatever the quality of the predictor and that are narrower than labeled-only intervals when the predictor is informative.

We ask a question about the *ceiling* of this recipe. Fix the labeled sample $(Y_i, f(X_i))_{i \le n}$, the unlabeled sample $(f(X_j))_{j \le N}$ and the predictor f; buy no labels, retrain nothing, add no unlabeled data. *Can a valid 95% interval for the population mean still be made shorter than the PPI++ interval, by how much, and under which conditions?* We separate three different sources of possible shortening, because each needs its own validity check: (a) a lower-variance point estimator (efficiency), (b) a better finite-sample critical value or variance estimate, and (c) less conservative constructions than CLT-based ones.

Our main object is (a). Within the information set above, the estimand has a semiparametric efficiency bound that depends on the data only through $E[Y \mid f]$ (Section 3); PPI++ attains it exactly when $E[Y \mid f]$ is affine in f. We therefore study estimators that learn a calibration map $g \approx E[Y \mid f]$

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

from the labels with cross-fitting and plug it into the PPI++ construction, and we measure on real and synthetic data, with known truth and thousands of replicates per cell, how much width this buys and whether coverage survives.

Answer in brief. (1) On the nine real datasets shipped with `ppi-python` the answer is essentially *no*: $E[Y \mid f]$ is nearly affine in the predictor, the efficiency ceiling lies within 0.5% of the PPI++ width on seven sets (for gene expression and census income it is 0.7% and 0.2% below the width of PPI++ without its clip), and a cross-fitted calibrated-score interval is wider than PPI++ in 24 of 26 primary cells, by up to 11% at $n = 50$, because learning the calibration map from the labels costs more than it gains. (2) Where $E[Y \mid f]$ is clearly nonlinear (synthetic populations) the calibrated interval can be shorter, by up to 21% in our grids, at coverage no better than PPI++'s in that cell (never below the ceiling), with the gain growing with n, N/n and signal-to-noise. (3) At small n the reference PPI++ interval itself undercovers (down to 0.84 on census income at $n = 50$); a jackknife-t correction restores coverage on most sets for 10–15% more width, so "shorter" claims at small n must be read against a reference that is not valid. (4) Under label–unlabeled distribution shift the unclipped power-tuned weight makes every calibrated or unclipped interval collapse (coverage 0.49 at $n = 400$ and 0.18 at $n = 1000$ on plankton) while the clipped package default survives.

Contributions. We (i) separate three sources of possible shortening and state the efficiency ceiling for the fixed-input setting with its assumptions (Proposition 1); (ii) measure it with 2000 paired replicates per cell on nine real datasets and a one-factor-at-a-time synthetic grid with known truth, reporting coverage with Wilson intervals and paired width ratios with bootstrap intervals; (iii) show that the measured gap to the ceiling is the plug-in cost of learning the calibration map and quantify it; (iv) document small-n undercoverage of the plug-in PPI++ interval and the width price of repairing it, and the shift failure of the unclipped weight. We do not propose a new estimator class: the calibrated estimator is the standard augmented estimator with a cross-fitted one-dimensional nuisance.

2 Related work

Prediction-powered inference. PPI [Angelopoulos et al., 2023a] debiases the mean prediction on unlabeled data with a rectifier estimated on the labeled data, and gives CLT intervals for means, quantiles and M-estimation targets. PPI++ [Angelopoulos et al., 2023b] adds a data-driven tuning weight λ on the predictions, so that the interval is never asymptotically wider than the labeled-only one. Cross-prediction-powered inference [Zrnic and Candès, 2023] treats the case where the predictor is trained on the same labels and uses cross-fitting. Active statistical inference [Zrnic and Candès, 2024] chooses which points to label from the model's uncertainty; stratified PPI [Fisch et al., 2024] uses strata of the prediction to allocate labels and to apply per-stratum rectifiers. Active selection needs control of the labeling mechanism and so is excluded by our fixed labeled set; the stratified idea (using f non-linearly) is the closest in spirit to the calibration we study. The reference software is `ppi-python` [Angelopoulos et al., 2024].

Semiparametric efficiency and debiased learning. Estimating a mean when the outcome is observed only for a random subset is the missing-data problem for which augmented inverse-probability-weighted estimators and their efficiency theory are classical [Robins et al., 1994, Tsiatis, 2006]. PPI with predictor f is the augmented estimator with the (generally misspecified) outcome regression f. Double/debiased machine learning [Chernozhukov et al., 2018] supplies the cross-fitting device that lets a nuisance function be learned from the same sample without bias. Our estimators are this construction with a one-dimensional nuisance $g(f)$; what we add is the fixed-input question and its quantification on the standard PPI datasets, not a new estimator class.

3 Setting, estimators and the efficiency ceiling

Setting. Records (X, Y) are i.i.d.; n are labeled and give (Y_i, f_i) with $f_i = f(X_i)$ for the fixed predictor f, and N more give only f_j. The target is $\theta = E[Y]$ (prevalence for binary Y) at level $1 - \alpha = 0.95$. The *only* inputs to an interval are the n pairs (Y_i, f_i) and the N scores f_j; covariates X other than through f are not used, even where a dataset has them. A map g of the scores learned from the labels is allowed only if learned inside the replicate with cross-fitting, so that the labels serve

once for the map and once for the rectifier. Write $\sigma^2 = \mathrm{Var}(Y)$, $\omega = N/(n+N)$, $g(f) = E[Y \mid f]$, $\rho^2 = \mathrm{Corr}(Y, f)^2$ and $R^2_{\mathrm{cal}} = \mathrm{Var}\, g(f)/\sigma^2 \geq \rho^2$, with equality iff g is affine.

PPI, PPI++ and the reference. For a score $s_i = t(f_i)$ and weight λ the estimate is $\hat{\theta}_\lambda = \frac{1}{n}\sum_{i\leq n}(Y_i - \lambda s_i) + \frac{\lambda}{N}\sum_{j\leq N} s_j$ with the Wald interval $\hat{\theta}_\lambda \pm z_{1-\alpha/2}\hat{s}_\lambda$, $\hat{s}^2_\lambda = \widehat{\mathrm{Var}}(Y - \lambda s)/n + \lambda^2\widehat{\mathrm{Var}}(s)/N$. $\lambda = 0$ is the classical interval (M1), $\lambda = 1$ is PPI [Angelopoulos et al., 2023a] (M2), and $\lambda = \hat{\lambda}$, the plug-in minimiser of the estimated variance, is PPI++ [Angelopoulos et al., 2023b]. Our reference (M3) is the `ppi-python` 0.2.3 default [Angelopoulos et al., 2024], which clips $\hat{\lambda}$ to $[0, 1]$ and uses a z quantile; it does not account for the estimation of $\hat{\lambda}$.

Efficiency ceiling. For a fixed function φ the estimator $\hat{\theta}_\varphi = \frac{1}{n}\sum_{i\leq n}(Y_i - \varphi(f_i)) + \frac{1}{N}\sum_{j\leq N}\varphi(f_j)$ is unbiased with asymptotic variance $V(\varphi) = \mathrm{Var}(Y-\varphi)/n + \mathrm{Var}(\varphi)/N$. PPI++ is $\varphi = \lambda f$ with the best λ, which gives $V_{\mathrm{PPI++}} = \sigma^2(1-\omega\rho^2)/n$. Because $\mathrm{Var}(Y-\varphi) = E\,\mathrm{Var}(Y \mid f) + \mathrm{Var}(g-\varphi)$, the minimiser over all φ is $\varphi^\star = \omega g + \mathrm{const}$, and

$$V^\star = \frac{E[\mathrm{Var}(Y \mid f)]}{n} + \frac{\mathrm{Var}\, g(f)}{n+N} = \frac{\sigma^2(1-\omega R^2_{\mathrm{cal}})}{n}. \tag{1}$$

Proposition 1. *Let $(Y_i, f_i)_{i\leq n}$ and $(f_j)_{j\leq N}$ be independent samples from the same law of f, $\sigma^2 < \infty$, and let $n, N \to \infty$ with ω fixed in $(0, 1)$. Then (i) the asymptotic width ratio of the best interval in the class $\{\hat{\theta}_\varphi\}$ to PPI++ (with unclipped λ) is $\sqrt{(1-\omega R^2_{\mathrm{cal}})/(1-\omega\rho^2)} \leq 1$, with equality iff g is affine in f (or $\omega \to 0$); (ii) $V^\star$ is the semiparametric efficiency bound for θ when labeling is completely at random and only f is used, so no regular estimator and no asymptotically valid Wald interval is narrower than $z_{1-\alpha/2}\sqrt{V^\star}$.*

The gain is thus bounded by the nonlinearity $R^2_{\mathrm{cal}} - \rho^2$ of $E[Y \mid f]$ and vanishes as $N/n \to 0$. Part (i) follows from the display above (Appendix A); part (ii) is the standard efficient-influence-function calculation for augmented estimators under missing-completely-at-random labels [Robins et al., 1994, Tsiatis, 2006], which we state without re-deriving. It describes the asymptotic regime only: it says nothing about finite-n coverage, about the cost of learning g, or about covariate shift between labeled and unlabeled data. If X were used, the floor could be lower still.

Calibrated-score PPI++ (M4). To approach $\varphi^\star$ with the labels alone we learn $\hat{g}$ from the labeled data by isotonic regression [Barlow et al., 1972], cross-fitted over $K = 5$ folds: labeled points are scored by the map fitted on the other folds, unlabeled points by the average of the five maps. The scores $s = \hat{g}(f)$ then enter the PPI++ interval with an unclipped, power-tuned $\hat{\lambda}$ and a z quantile. The isotonic map was chosen, before any final run, among isotonic, spline and binned maps by the smallest mean log paired width ratio to M3 on 45 development synthetic cells (seed space disjoint from the final runs; `code/results/frozen_choices.json`). Because $\hat{g}$ is estimated from n labels, M4 cannot reach $V^\star$ at finite n; the excess is the *plug-in cost*.

Small-sample corrections (M5). M5 replaces the CLT critical value or the variance estimate. For M3 it uses a delete-one jackknife standard error (with $\hat{\lambda}$ re-estimated in each leave-one-out fit) and a t_{n-1} quantile; for M4 it uses the t_{n-1} quantile only. These are the finite-sample calibration route (b) of the introduction; they lengthened the intervals in every cell here.

4 Experimental design

Data and sampling model. Real data are the nine example sets of `ppi-python` 0.2.3 [Angelopoulos et al., 2024] (labels Y and predictions f): galaxies (spiral indicator; Willett et al. 2013), forest, AlphaFold, gene expression, census income (income, continuous; Ding et al. 2021), ballots, census healthcare and education (appendix), and plankton (distribution-shift stress test). Each dataset's pool is the population and θ is the pool mean of Y. Labeled and unlabeled records are drawn independently and i.i.d. *with replacement* from the pool, so the fraction of the pool used does not enter, with $n \in \{50, 100, 200, 400, 1000\}$ capped at pool$/5$ (forest and ballots: $n \leq 200$) and $N = 20n$; a second sweep varies $N \in \{n, 5n, 20n, 20000\}$. For ballots, whose real unlabeled split has no labels,

Table 1: Methods compared. All use only the labeled (Y, f) and the unlabeled f.

	Method	Point estimate	Information used	Critical value
M1	classical	mean of Y on the n labeled	labels only	z
M2	PPI	$\bar{f}_{\mathrm{unl}} + \overline{(Y-f)}_{\mathrm{lab}}$	(Y, f) labeled, f unlabeled	z
M3	PPI++ (`ppi_py` 0.2.3, reference)	power-tuned $\hat{\lambda} \in [0, 1]$	same as M2	z
M4	isotonic-calibrated PPI++	PPI++ on a score $\hat{g}(f)$ learned on the labels, 5-fold cross-fitted	same as M2 (covariates not used)	z
M5	small-n corrections	M3 with delete-one jackknife s.e.; M4 with t_{n-1}	same as M2	t_{n-1}

the pool is its 1027 labeled records. For plankton the labeled records come from the 2013 pool and the unlabeled predictions from 2014, with θ the mean of the 2014 labels; the control draws both from 2013. A forest ablation draws labeled records without replacement. Synthetic populations with known truth (Appendix B) vary the nonlinearity κ of $E[Y \mid f]$ ($\tanh$, quadratic, and a binary outcome with an overconfident classifier), n, N/n, the signal-to-noise ratio and the noise tail.

Replication and uncertainty. Every cell uses $R = 2000$ replicates, and all methods see the same draws within a replicate, so width comparisons are paired. Coverage is reported with its Wilson interval [Wilson, 1927]; the Monte-Carlo standard error is ≤ 0.0049 near 0.95. Width is the paired geometric-mean ratio to M3 with a 95% percentile bootstrap interval over replicates. A cell is flagged below nominal (bold in tables) when the Wilson upper bound is < 0.95. We make a coverage claim only for a pattern across neighbouring n or datasets, not for single cells; no cell-level p-values are computed, and the 26 primary real cells and the synthetic cells are not corrected for multiplicity. Seeds are derived from (dataset, n, N) in a final space disjoint from the development space used to choose M4 and M5. Tables 1 and the reproduction notes list methods, seeds, commands and run times.

5 Results

5.1 On the real datasets the fixed-input ceiling is almost reached by PPI++

Table 2 gives, for each dataset, the population quantities that enter the ceiling (1). $E[Y \mid f]$ is nearly affine in f on all of them: $|R^2_{\mathrm{cal}} - \rho^2| < 0.01$ everywhere (the cross-fitted estimate can fall slightly below ρ^2, as on forest), so the predicted asymptotic width ratio (a floor estimated with cross-fitted isotonic maps, hence occasionally slightly above 1, as on forest and ballots) of a calibrated interval to M3 is within $\pm 0.5\%$ of 1 for seven of the nine sets, 0.983 for census income and 0.964 for gene expression. Most of these two gains is the clip, not calibration: against *unclipped* PPI++ the ratios are 0.998 and 0.993, because there the optimal $\hat{\lambda}$ exceeds one and M3 clips it.

Figure 1 shows the measured effect of calibration. M4 is wider than M3 in 24 of the 26 primary cells (paired ratio 1.02–1.11 at $n = 50$, 1.006–1.046 at $n = 200$, 0.995–1.025 at $n = 400$ and 0.979–1.009 at $n = 1000$). The two exceptions are gene expression at $n = 400$ and 1000 (ratios 0.995 and 0.979); unclipped PPI++ with an affine score (`ppipp_nc`) gives 0.979 and 0.974 in the same cells, so the gain there is the removal of the clip, not calibration. M2 (PPI, $\lambda = 1$) is within $\pm 2.5\%$ of M3 everywhere, and M1 is 1.2 to 3.7 times wider (Appendix Table 9). Table 3 summarises the regimes. Hence *no calibrated-score interval of the kind studied is appreciably shorter than PPI++ on these datasets*, consistent with the ceiling; the two exceptions below are the removal of the clip.

The excess width of M4 over the ceiling is the plug-in cost of learning $\hat{g}$ (Figure 2): measured/predicted -1 is 1.4–12.6% at $n = 50$, 1.7–4.3% at $n = 400$ and 0.8–2.4% at $n = 1000$ (smallest on ballots), and on galaxies and forest it grows with N/n, as expected if a larger unlabeled set raises the optimal weight on $\hat{g}$ (we did not test this; on gene expression and census income it does not grow) (galaxies, $n = 100$: 1.020, 1.052 and 1.059 for $N = n$, $20n$ and $200n$).

5.2 When shortening is possible: nonlinear $E[Y \mid f]$

On synthetic populations where $E[Y \mid f] = \sqrt{1-\kappa}\, f + \sqrt{\kappa}\, h(f)$ the ceiling is lower and M4 reaches part of it (Figure 3). With $\kappa = 0$ (affine) M4 is 6% wider than M3 at $n = 100$, $N = 20n$, the

Table 2: Population quantities behind the ceiling (1) for each dataset (pool values); the last two columns are the asymptotic width ratio of an isotonic-calibrated interval to M3 without estimation cost, at $N = 20n$.

Dataset	Pool size	$\mathrm{Var}(Y)$	ρ^2	R^2_{cal}	floor ratio, $n = 100$	floor ratio, $n = 400$
Galaxies	16 743	0.1921	0.549	0.547	1.001	1.001
Forest	1 596	0.1286	0.289	0.281	1.005	1.005
AlphaFold	10 802	0.1459	0.664	0.665	0.999	0.999
Gene expression	61 150	19.28	0.928	0.930	0.964	0.964
Census income	380 091	5.758×10^9	0.369	0.372	0.983	0.983
Census healthcare	318 215	0.2208	0.500	0.500	1.000	1.000
Census education	318 215	19.23	0.392	0.392	1.000	1.000
Ballots	1 027	0.2336	0.963	0.963	1.004	–
Plankton	421 238	0.1159	0.699	0.699	1.000	1.000

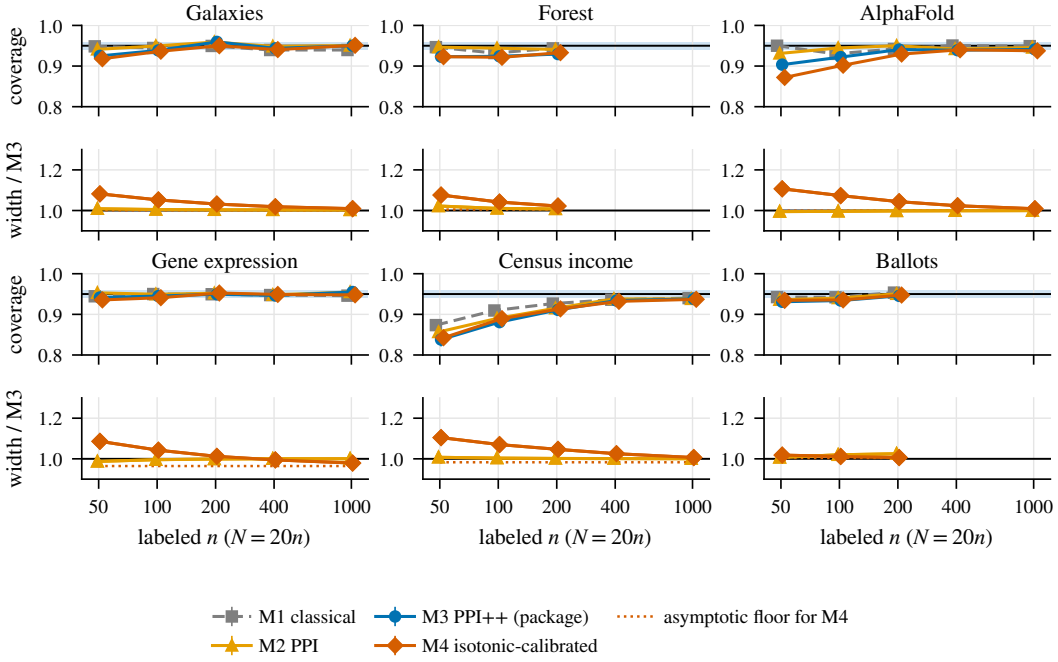


Figure 1: Coverage (top rows, Wilson 95% intervals; shaded band: range of a nominally covering method's Monte-Carlo estimate) and paired width ratio to M3 (bottom rows; 95% bootstrap interval; dotted: asymptotic ratio for M4 from population quantities) on six real datasets, $N = 20n$, $R = 2000$.

pure cost of cross-fitting; an affine cross-fitted map is similar (1.03). For a saturating $h = \tanh$ the ratio is 0.951 at $\kappa = 0.75$ against a ceiling of 0.897. For a quadratic h (a U-shape that no affine score can use) the ratio is 0.912 at $\kappa = 0.75$ (ceiling 0.773) and 0.930 at $\kappa = 0.5$ (ceiling 0.830), and falls to 0.86 at $n = 1000$ and 0.79 at SNR $= 4$; the measured ratio is above the ceiling in every one of the 45 synthetic cells (mean log gap 0.07, range 0.01–0.27; log-correlation 0.88 between predicted and measured ratios). For the binary outcome with a miscalibrated classifier M4 is never narrower than M3 except at $n = 400$, $\kappa = 0$ (ratio 0.989), because $E[Y \mid f]$ is smooth and close to affine in the score's range. Isotonic regression cannot represent a U-shape: on the quadratic population the final-run spline calibrator (not selected as M4) reaches 0.83 at $\kappa = 1$ against the ceiling 0.73, while isotonic gives 0.95 (Appendix B; the choice of M4 was fixed before these runs, so this is a post-hoc ablation). With $\kappa > 0$ the gain is therefore bounded by the nonlinearity, larger with more labels (plug-in cost shrinks), more unlabeled data and cleaner labels.

5.3 Validity at small n: the plug-in interval undercovers, and the cure costs width

M3 undercovers at small n in a consistent pattern (Table 3, Appendix Table 8 for $n = 100, 400$ and `tab_real_full.csv` for all cells; Figures 1 and 4): at $n = 50$ the coverage is 0.904 (AlphaFold),

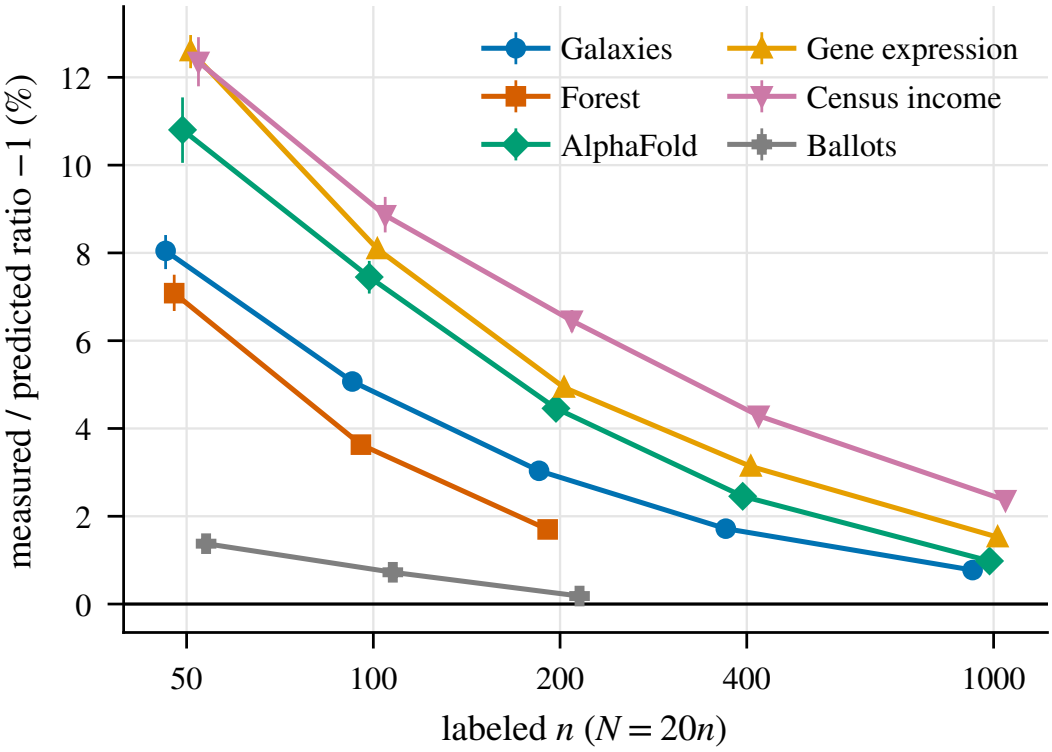


Figure 2: Plug-in cost of M4: measured over predicted width ratio minus one, versus n ($N = 20n$).

Table 3: Regimes by dataset (real rows: 36 cells in all; synthetic rows: the 23 κ- and n-sweep cells at SNR 1): number of cells where M4 is narrower, indistinguishable or wider than M3 (paired 95% interval), the range of the ratio, and the number of cells below nominal for M3 and M4.

Cells	#	M4 narrower	indistinguishable	M4 wider	smallest ratio	largest ratio	M3 undercovers	M4 undercovers
Galaxies	5	0	0	5	1.009	1.082	2	2
Forest	3	0	0	3	1.022	1.076	3	3
AlphaFold	5	0	0	5	1.009	1.107	3	5
Gene expression	5	2	0	3	0.979	1.086	0	1
Census income	5	0	0	5	1.006	1.104	5	5
Census healthcare	5	0	0	5	1.009	1.073	1	2
Census education	5	0	0	5	1.013	1.092	3	2
Ballots	3	0	0	3	1.006	1.018	2	2
synthetic, $\kappa = 0$	3	0	0	3	1.017	1.060	1	0
synthetic, tanh/quad, $\kappa > 0$	16	15	1	0	0.864	0.998	4	3
synthetic, binary, $\kappa > 0$	4	0	0	4	1.015	1.018	3	3

0.838 (census income), 0.923 (forest) and 0.925 (galaxies), and in 15 of the 26 primary cells the Wilson upper bound is below 0.95. M4 has lower coverage on AlphaFold ($n = 100$: 0.902 against 0.922) and galaxies and comparable coverage elsewhere (all cells: `tab_real_full.csv` in the delivered tables). The delete-one jackknife with t_{n-1} (M5 on M3) raises these to 0.954, 0.868, 0.942 and 0.949 at $n = 50$ for 10–15% more width (2–4% at $n = 200$); t_{n-1} alone on M4 helps less (AlphaFold $n = 50$: 0.880). Census income stays below 0.95 up to $n = 400$ for every method, including the classical z interval (0.874 at $n = 50$); we suspect skewed income labels but did not test this. A bootstrap ablation (Appendix Table 4) shows that a percentile bootstrap of PPI++ does not repair coverage, while a studentized bootstrap does so in most cells except census income, at 2–24% greater width (14–49% on census income, which it does not repair). The conformal baseline of `ppi-python` is the whole real line unless $n \geq N/\alpha - 1$, which never holds at $N = 20n$, so it gives no finite interval in our design.

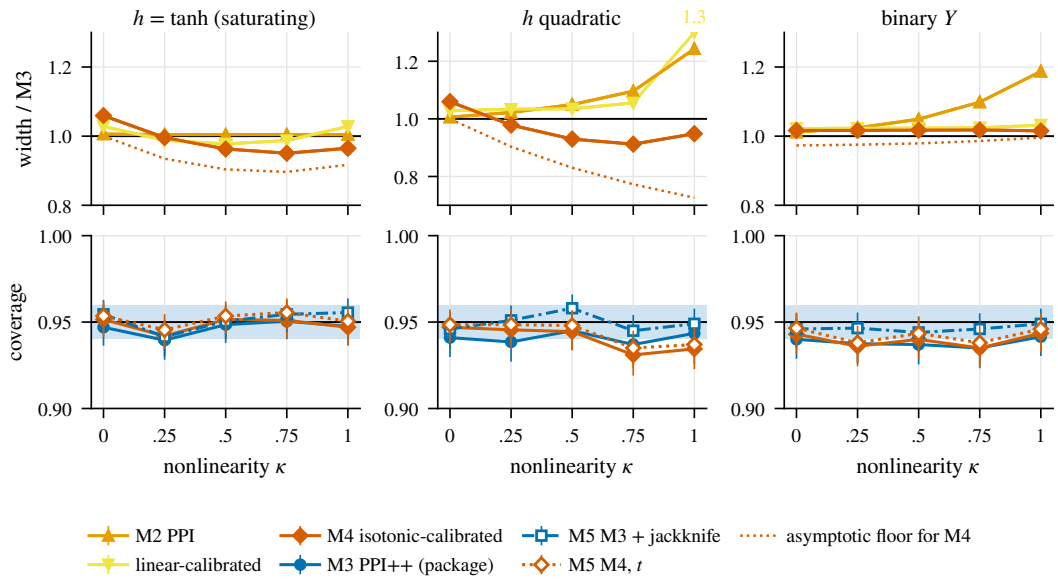


Figure 3: Synthetic populations: paired width ratio to M3 and coverage versus the nonlinearity κ ($n = 100$, $N = 20n$, SNR $= 1$, $R = 2000$). Dotted: asymptotic ceiling for M4.

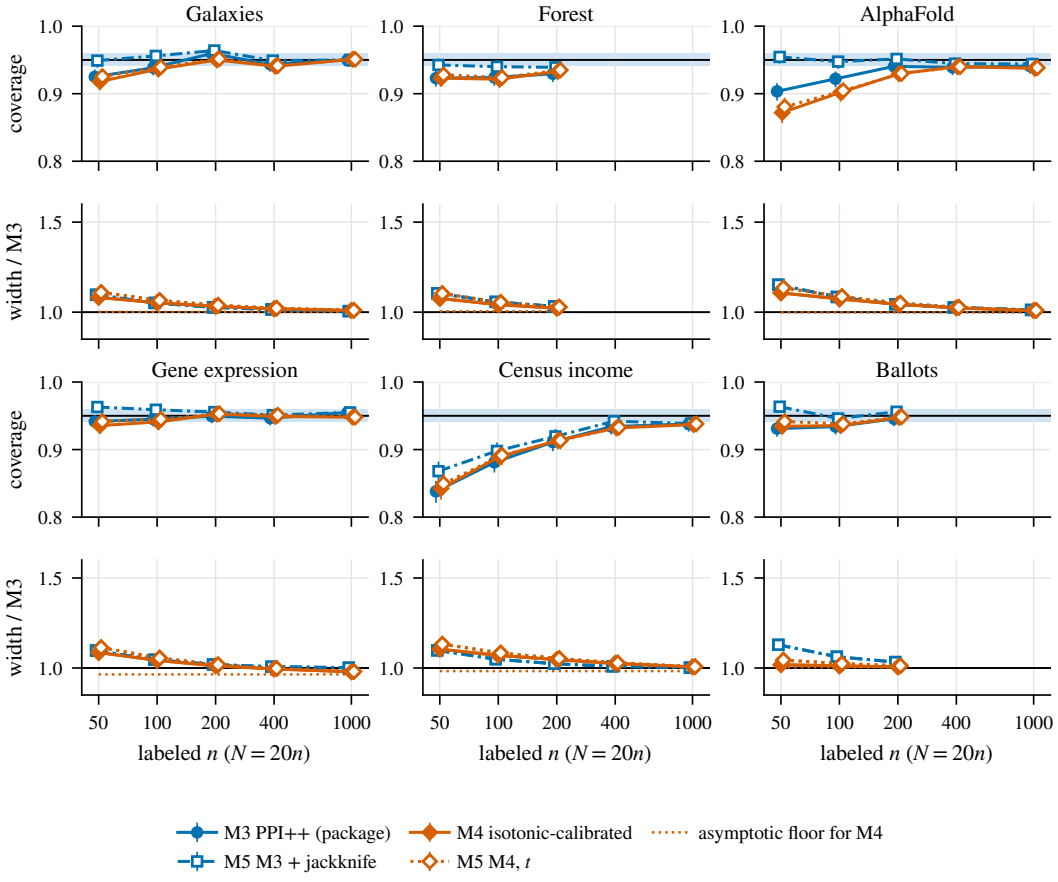


Figure 4: Small-n corrections on the six primary real datasets: M3, M5 on M3 (jackknife, t), M4 and M5 on M4 (t). Layout as in Figure 1.

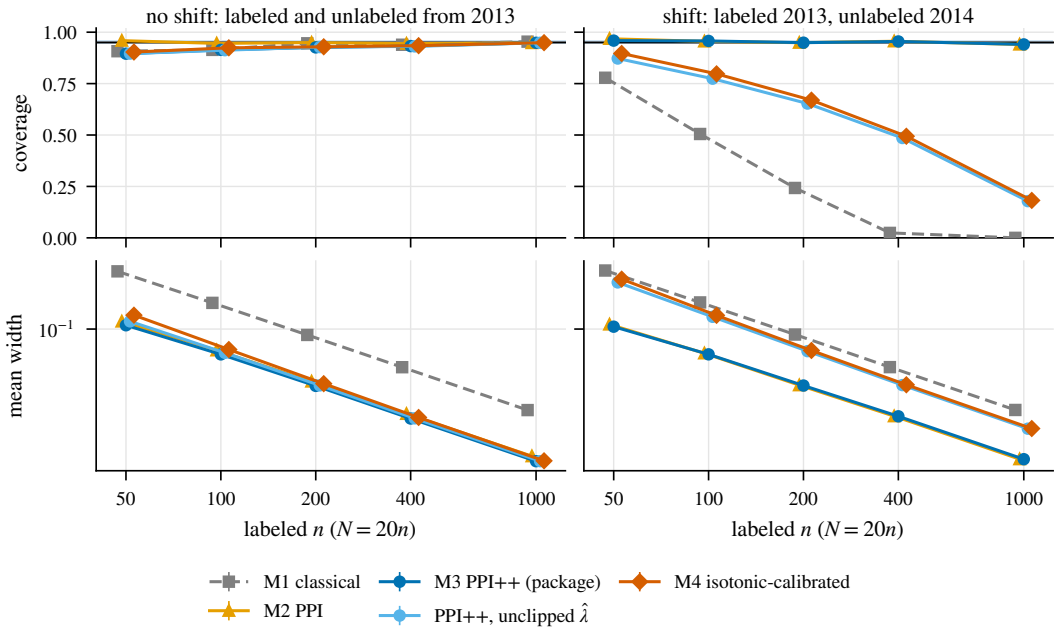


Figure 5: Plankton: labeled 2013 versus unlabeled 2014 (shift) against a within-2013 control; coverage versus n ($R = 2000$).

5.4 Distribution shift breaks the optimal-weight intervals

On plankton, labeled records from 2013 and unlabeled predictions from 2014 differ in distribution (mean Y 0.134 versus 0.071). The classical interval's coverage falls to 0.00 at $n = 1000$ (bias), while M2, M3 and the jackknife M5 keep 0.94–0.97 (Figure 5). Every variant with an unclipped or calibrated $\hat{\lambda}$, M4 included, degrades with n (M4 coverage 0.90, 0.80, 0.67, 0.49, 0.18 at $n = 50, 100, 200, 400, 1000$; at $n = 400$ unclipped PPI++ 0.49, within-2013 control 0.93–0.94). A direct check (`code/src/shift_lambda_check.py`) shows why: under shift the variance of the unlabeled (2014) scores is smaller than that of the 2013 scores (0.062 versus 0.114), so $\hat{\lambda} \approx 1.40$ in 82–100% of draws (control: 0.80; $R = 300$ draws per n), which puts negative weight on the labeled-minus-unlabeled prediction contrast that carries the shift; M3 survives only because the package clips $\hat{\lambda}$ at 1. An affine cross-fitted score fails equally, so calibration itself is not the cause.

6 Discussion and limitations

What the evidence supports. With labels, unlabeled data and predictor fixed, the room for shorter valid CLT intervals for a mean is the nonlinearity of $E[Y \mid f]$ beyond its best affine fit, plus whatever PPI++'s clip of $\hat{\lambda}$ discards. On the real datasets studied the first is below 1% of the width, the second matters only for gene expression and census income, and the estimation cost of a calibration map outweighs it at every $n \leq 200$; only the synthetic nonlinear populations show a benefit, and it stayed above the ceiling in all 45 cells. We therefore read the outcome as *conditional*: possible gains exist but need a predictor whose conditional mean is clearly non-affine in its own output, e.g. a systematically miscalibrated or non-monotone score, and enough labels to learn the map (hundreds). Absent that, PPI++ is already near the ceiling.

Limitations. (a) *Estimand and information set:* only the population mean (prevalence for binary Y); quantiles and regression coefficients are not studied. Covariates X are not used; the ceiling would be lower if they were, and we did not measure that headroom on the census sets. (b) *Sampling model:* the pool is the population and records are drawn with replacement, so finite-population effects of the original data protocols are not studied; a without-replacement forest ablation shows no clear difference at $n = 50$ and 100 and higher coverage at $n = 200$ (0.952 against 0.930), as the

finite-population correction would give (Appendix Table 5). Cells at different n share one pool and are not independent. (c) *Methods:* one calibrator class (isotonic, $K = 5$) was frozen before the final runs; it cannot fit non-monotone maps, and the spline ablation shows a larger gain in that case, but spline was not pre-selected. Learning $\hat{g}$ also reduces effective sample size at small n; we do not claim the best possible estimator of g. (d) *Evidence:* the Monte-Carlo standard error of a coverage is about 0.005 (95% half-width about 0.01); we report cells below nominal rather than testing them and do not correct for multiplicity, so single-cell shortfalls should not be over-read, only the consistent patterns across n and datasets. The cause of census-income undercoverage is untested. (e) *Theory:* Proposition 1 is asymptotic; it does not describe finite-n coverage or the plug-in cost, which we only measure. (f) *Design:* most cells use $N = 20n$; the N/n sweep covers four real sets at $n = 100, 400$. The conformal baseline is not simulated (it is infinite here). Ballots have no labeled unlabeled split, so its pool is the labeled set. (g) *Shift:* the plankton result is one real shift; we explain it through $\hat{\lambda}$ but did not test remedies.

Broader impact. Intervals that undercover silently mislead downstream science, which is the risk behind the small-n and shift findings; we release only analysis code on public data.

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A Efficiency ceiling: derivation and scope

Let φ be a fixed measurable function with $E\varphi(f)^2 < \infty$. Then $\hat{\theta}_\varphi = \bar{Y}_{\mathrm{lab}} - \bar{\varphi}_{\mathrm{lab}} + \bar{\varphi}_{\mathrm{unl}}$ is unbiased for $E[Y]$ because labeled and unlabeled scores have the same law, and the three terms are independent given the design, so $\mathrm{Var}\,\hat{\theta}_\varphi = \mathrm{Var}(Y - \varphi(f))/n + \mathrm{Var}(\varphi(f))/N = V(\varphi)$. Decompose $Y - \varphi = (Y-g)+(g-\varphi)$, with $E[(Y-g)(g-\varphi)] = 0$: $V(\varphi) = E\mathrm{Var}(Y \mid f)/n + \mathrm{Var}(g-\varphi)/n + \mathrm{Var}(\varphi)/N$. Write $\varphi = cg + h$ with h uncorrelated with g; then $\mathrm{Var}(g-\varphi) = (1-c)^2\mathrm{Var}\,g + \mathrm{Var}\,h$ and $\mathrm{Var}\,\varphi = c^2\mathrm{Var}\,g + \mathrm{Var}\,h$, so $V(\varphi) = E\mathrm{Var}(Y \mid f)/n + \mathrm{Var}\,g\,[(1-c)^2/n + c^2/N] + \mathrm{Var}\,h\,(1/n + 1/N)$. The last term is minimised at $h = 0$ and the bracket at $c = \omega = N/(n+N)$, where it equals $1/(n+N)$. Hence $V^\star = E\mathrm{Var}(Y \mid f)/n + \mathrm{Var}\,g/(n+N)$, and with $E\mathrm{Var}(Y \mid f) = \sigma^2(1 - R^2_{\mathrm{cal}})$ this is $\sigma^2(1 - \omega R^2_{\mathrm{cal}})/n$, the display (1). For $\varphi = \lambda f$ the same algebra with g replaced by its best linear predictor gives $\sigma^2(1 - \omega\rho^2)/n$ at the optimal $\lambda = \mathrm{Cov}(Y, f)/\{(1 + n/N)\mathrm{Var}(f)\}$. The ratio of (1) to this is $(1 - \omega R^2_{\mathrm{cal}})/(1 - \omega\rho^2) \le 1$ because $R^2_{\mathrm{cal}} \ge \rho^2$, with equality iff g is affine. The efficiency-bound claim is the standard augmented-IPW result for a missing-completely-at-random outcome observed on a fraction $n/(n+N)$ of records with the observed-data information restricted to f [Robins et al., 1994, Tsiatis, 2006]. The Wald interval built on the plug-in $\hat{V}$ is asymptotically valid under the stated moment conditions as $n, N \to \infty$; cross-fitting $\hat{g}$ preserves this when $\hat{g}$ converges in L^2 to a limit map [Chernozhukov et al., 2018]. The cost of learning $\hat{g}$ (the plug-in cost) is lower order in n but is what the experiments measure; no formula for it is proved here.

B Synthetic populations

$f \sim N(0,1)$ is the predictor output. Gaussian family: $Y = g_\kappa(f) + \varepsilon/\sqrt{\mathrm{SNR}}$ with $g_\kappa = \sqrt{1-\kappa}\,f + \sqrt{\kappa}\,h(f)$, $\varepsilon \sim N(0,1)$ (or Student-t_3 rescaled to unit variance for the heavy-tail cells), $h(f) = \tanh(2f)/0.7$ or $h(f) = (f^2 - 1)/\sqrt{2}$, so $\theta = 0$. Binary family: $f_{\mathrm{prob}} = \mathrm{sigmoid}(3z)$ with $z \sim N(0, \mathrm{SNR}^2)$ and $Y \sim \mathrm{Bernoulli}(\mathrm{sigmoid}(a\,\mathrm{logit}\,f_{\mathrm{prob}} - 1))$ with temperature $a = 1 - 0.8\kappa$ ($a = 1$ is a calibrated classifier); θ is computed from 2×10^7 draws. The grid is one factor at a time around the base cells ($\tanh$ or quadratic, $\kappa = 0.5$, SNR 1, $n = 100$, $N = 20n$): $\kappa \in \{0, \ldots, 1\}$ for each family, $n \in \{50, \ldots, 1000\}$, $N \in \{n, 5n, 20n, 20000\}$, SNR $\in \{0.25, \ldots, 4\}$, heavy tails, 45 cells with $R = 2000$ replicates each (`code/src/grids.py`). Figure 8 gives the n, N and SNR panels.

C Additional results

The spline calibrator (not the frozen M4) on the quadratic family, $n = 100$, $N = 20n$: paired width ratio to M3 1.086, 0.996, 0.933, 0.882, 0.827 for $\kappa = 0, 0.25, 0.5, 0.75, 1$ against ceilings 1.000, 0.903, 0.830, 0.773, 0.726 (`code/results/theory_check_synth_final.csv`); isotonic gives 1.060, 0.979, 0.930, 0.912, 0.948.

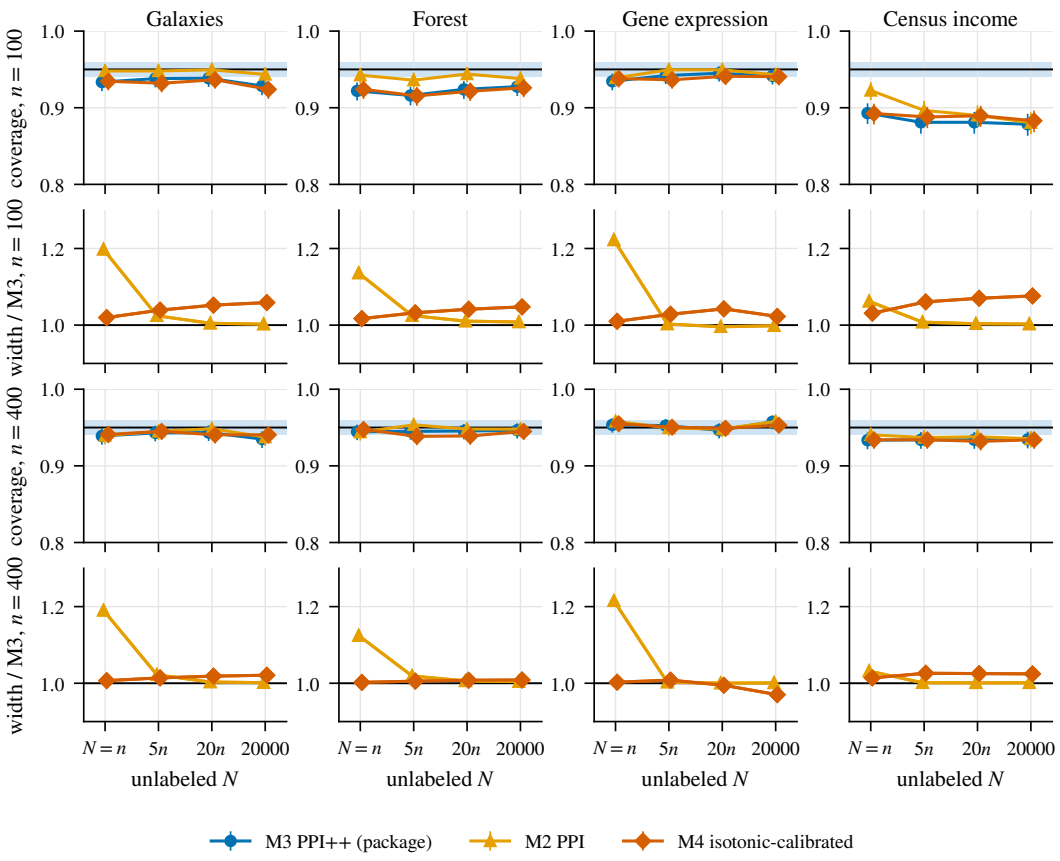


Figure 6: Dependence on the unlabeled sample size $N \in \{n, 5n, 20n, 20000\}$ at $n = 100$ and 400.

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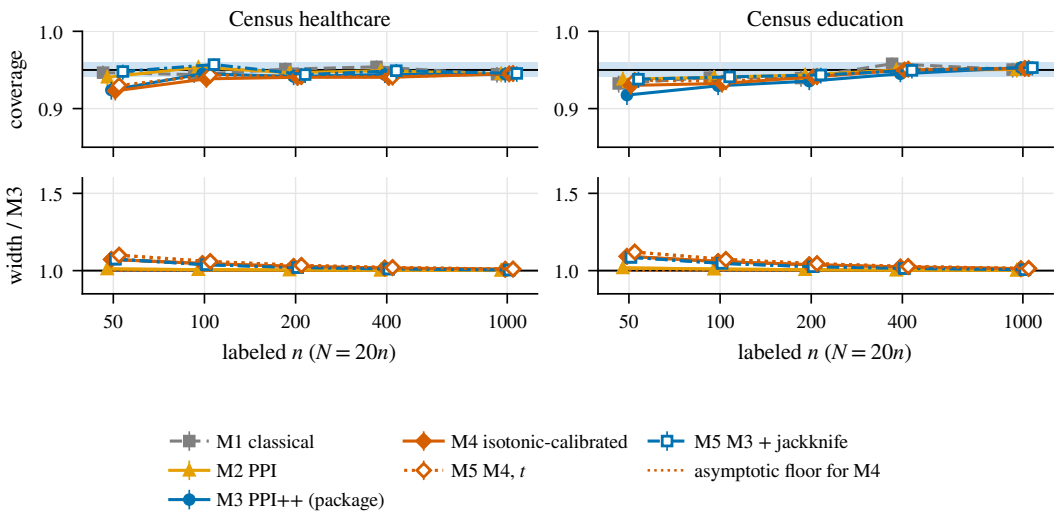


Figure 7: Census healthcare and education, $N = 20n$.

Justification: Section 6 lists limitations: mean estimand only, covariates unused, resampling-with-replacement model, one frozen calibrator, no multiplicity correction, asymptotic theory only.

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Table 4: Bootstrap ablation ($R = 1000$): unclipped PPI++ (`ppipp_nc`) against percentile and studentized bootstrap.

Dataset	n	N	PPI++ cov.	boot. pct cov.	boot. pct width / PPI++	boot. t cov.	boot. t width / PPI++
AlphaFold	50	250	**0.907**	**0.912**	1.141	0.943	1.201
AlphaFold	50	1000	**0.880**	**0.871**	1.209	**0.914**	1.235
AlphaFold	100	500	**0.934**	**0.928**	1.075	0.962	1.103
AlphaFold	100	2000	**0.915**	**0.915**	1.118	0.959	1.127
AlphaFold	200	1000	0.943	0.946	1.036	0.950	1.053
AlphaFold	200	4000	0.945	0.947	1.064	0.956	1.070
Ballots	50	250	**0.934**	**0.936**	1.095	0.958	1.105
Ballots	50	1000	**0.935**	0.953	1.208	0.946	1.194
Ballots	100	500	0.945	0.948	1.041	0.956	1.045
Ballots	100	2000	0.937	0.941	1.084	**0.931**	1.106
Ballots	200	1000	0.956	0.951	1.014	0.956	1.017
Ballots	200	4000	0.952	0.950	1.036	0.944	1.052
Census income	50	250	**0.838**	**0.844**	1.002	**0.903**	1.427
Census income	50	1000	**0.810**	**0.819**	1.046	**0.885**	1.487
Census income	100	500	**0.878**	**0.880**	0.999	**0.913**	1.269
Census income	100	2000	**0.867**	**0.866**	1.009	**0.896**	1.295
Census income	200	1000	**0.916**	**0.914**	0.991	**0.931**	1.139
Census income	200	4000	**0.902**	**0.907**	0.999	**0.931**	1.160
Forest	50	250	**0.910**	**0.908**	1.068	**0.927**	1.180
Forest	50	1000	**0.926**	**0.919**	1.106	0.942	1.184
Forest	100	500	**0.922**	**0.924**	1.032	0.966	1.094
Forest	100	2000	**0.912**	**0.920**	1.052	0.962	1.103
Forest	200	1000	0.949	0.946	1.013	0.956	1.043
Forest	200	4000	**0.933**	**0.933**	1.023	0.952	1.048
Galaxies	50	250	**0.926**	**0.927**	1.057	0.951	1.121
Galaxies	50	1000	**0.909**	**0.905**	1.069	0.949	1.125
Galaxies	100	500	0.949	0.944	1.026	0.961	1.054
Galaxies	100	2000	**0.930**	**0.930**	1.037	0.943	1.061
Galaxies	200	1000	0.937	**0.936**	1.006	0.938	1.018
Galaxies	200	4000	0.943	0.943	1.014	0.946	1.026
Gene expression	50	250	**0.931**	0.941	1.095	0.959	1.092
Gene expression	50	1000	**0.931**	0.949	1.175	0.959	1.120
Gene expression	100	500	0.957	0.964	1.044	0.962	1.043
Gene expression	100	2000	0.938	0.947	1.085	0.955	1.059
Gene expression	200	1000	0.938	0.939	1.020	0.948	1.020
Gene expression	200	4000	0.947	0.944	1.047	0.956	1.033

Table 5: Forest, labeled draws with versus without replacement.

n	M3 cov., with repl.	M4 cov., with repl.	M4 / M3, with repl.	M3 cov., without repl.	M4 cov., without repl.	M4 / M3, without repl.
50	**0.923**	**0.923**	1.076	**0.908**	**0.906**	1.079
100	**0.924**	**0.921**	1.041	**0.936**	**0.930**	1.043
200	**0.929**	**0.933**	1.022	0.952	0.949	1.025

Justification: Proposition 1 states its assumptions; part (i) is proved in Appendix A, part (ii) is the standard efficiency-bound result, cited and not re-derived (stated as such).

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Table 6: Intervals on one realisation at native dataset sizes (10% labeled, 90% unlabeled, seed 0); descriptive only, no coverage claim.

Dataset	n	N	Method	Interval	Width	Covers θ
Galaxies	1674	15069	M1 classical	[0.2447, 0.2870]	0.04234	True
Galaxies	1674	15069	M2 PPI	[0.2518, 0.2823]	0.03045	True
Galaxies	1674	15069	ppipp	[0.2518, 0.2820]	0.03020	True
Galaxies	1674	15069	M4 isotonic-calibrated	[0.2509, 0.2813]	0.03041	True
Galaxies	1674	15069	binned-calibrated	[0.2517, 0.2823]	0.03058	True
Galaxies	1674	15069	spline-calibrated	[0.2518, 0.2821]	0.03031	True
Forest	159	1437	M1 classical	[0.1387, 0.2638]	0.1250	True
Forest	159	1437	M2 PPI	[0.1411, 0.2525]	0.1114	True
Forest	159	1437	ppipp	[0.1422, 0.2528]	0.1106	True
Forest	159	1437	M4 isotonic-calibrated	[0.1364, 0.2516]	0.1152	True
Forest	159	1437	binned-calibrated	[0.1319, 0.2510]	0.1191	True
Forest	159	1437	spline-calibrated	[0.1357, 0.2518]	0.1161	True
AlphaFold	1080	9722	M1 classical	[0.1585, 0.2045]	0.04599	True
AlphaFold	1080	9722	M2 PPI	[0.1711, 0.2011]	0.03000	True
AlphaFold	1080	9722	ppipp	[0.1708, 0.2007]	0.02988	True
AlphaFold	1080	9722	M4 isotonic-calibrated	[0.1699, 0.2001]	0.03021	True
AlphaFold	1080	9722	binned-calibrated	[0.1686, 0.1995]	0.03088	True
AlphaFold	1080	9722	spline-calibrated	[0.1703, 0.2004]	0.03010	True
Gene expression	6115	55035	M1 classical	[6.637, 6.859]	0.2217	True
Gene expression	6115	55035	M2 PPI	[6.738, 6.829]	0.09035	True
Gene expression	6115	55035	ppipp	[6.738, 6.829]	0.09035	True
Gene expression	6115	55035	M4 isotonic-calibrated	[6.742, 6.832]	0.08982	True
Gene expression	6115	55035	binned-calibrated	[6.746, 6.838]	0.09231	True
Gene expression	6115	55035	spline-calibrated	[6.742, 6.832]	0.08962	True
Census income	38009	342082	M1 classical	[42062, 43574]	1512	True
Census income	38009	342082	M2 PPI	[42278, 43520]	1242	True
Census income	38009	342082	ppipp	[42278, 43520]	1242	True
Census income	38009	342082	M4 isotonic-calibrated	[42230, 43458]	1228	True
Census income	38009	342082	binned-calibrated	[42400, 43659]	1259	True
Census income	38009	342082	spline-calibrated	[42225, 43453]	1229	True
Census healthcare	31821	286394	M1 classical	[0.6634, 0.6738]	0.01034	True
Census healthcare	31821	286394	M2 PPI	[0.6651, 0.6728]	0.007706	True
Census healthcare	31821	286394	ppipp	[0.6651, 0.6728]	0.007659	True
Census healthcare	31821	286394	M4 isotonic-calibrated	[0.6652, 0.6729]	0.007656	True
Census healthcare	31821	286394	binned-calibrated	[0.6651, 0.6728]	0.007684	True
Census healthcare	31821	286394	spline-calibrated	[0.6652, 0.6728]	0.007657	True
Census education	31821	286394	M1 classical	[17.76, 17.86]	0.09654	True
Census education	31821	286394	M2 PPI	[17.78, 17.85]	0.07835	True
Census education	31821	286394	ppipp	[17.78, 17.85]	0.07790	True
Census education	31821	286394	M4 isotonic-calibrated	[17.78, 17.85]	0.07797	True
Census education	31821	286394	binned-calibrated	[17.78, 17.86]	0.07850	True
Census education	31821	286394	spline-calibrated	[17.78, 17.85]	0.07790	True
Ballots	1027	78150	M1 classical	[0.5985, 0.6576]	0.05915	NaN
Ballots	1027	78150	M2 PPI	[0.6145, 0.6278]	0.01330	NaN
Ballots	1027	78150	ppipp	[0.6149, 0.6280]	0.01313	NaN
Ballots	1027	78150	M4 isotonic-calibrated	[0.6149, 0.6281]	0.01320	NaN
Ballots	1027	78150	binned-calibrated	[0.6149, 0.6281]	0.01320	NaN
Ballots	1027	78150	spline-calibrated	[0.6149, 0.6281]	0.01320	NaN
Plankton	421238	329832	M1 classical	[0.1327, 0.1348]	0.002056	False
Plankton	421238	329832	M2 PPI	[0.06890, 0.07097]	0.002069	False
Plankton	421238	329832	ppipp	[0.1038, 0.1054]	0.001582	False
Plankton	421238	329832	M4 isotonic-calibrated	[0.1038, 0.1054]	0.001582	False
Plankton	421238	329832	binned-calibrated	[0.1038, 0.1054]	0.001582	False
Plankton	421238	329832	spline-calibrated	[0.1038, 0.1054]	0.001582	False

Justification: Sections 3 and 4 and Appendix B define estimators, sampling model, grids, replicates, seeds and the choice of M4 and M5 on development seeds; the reproduction notes (README) list versions and commands.

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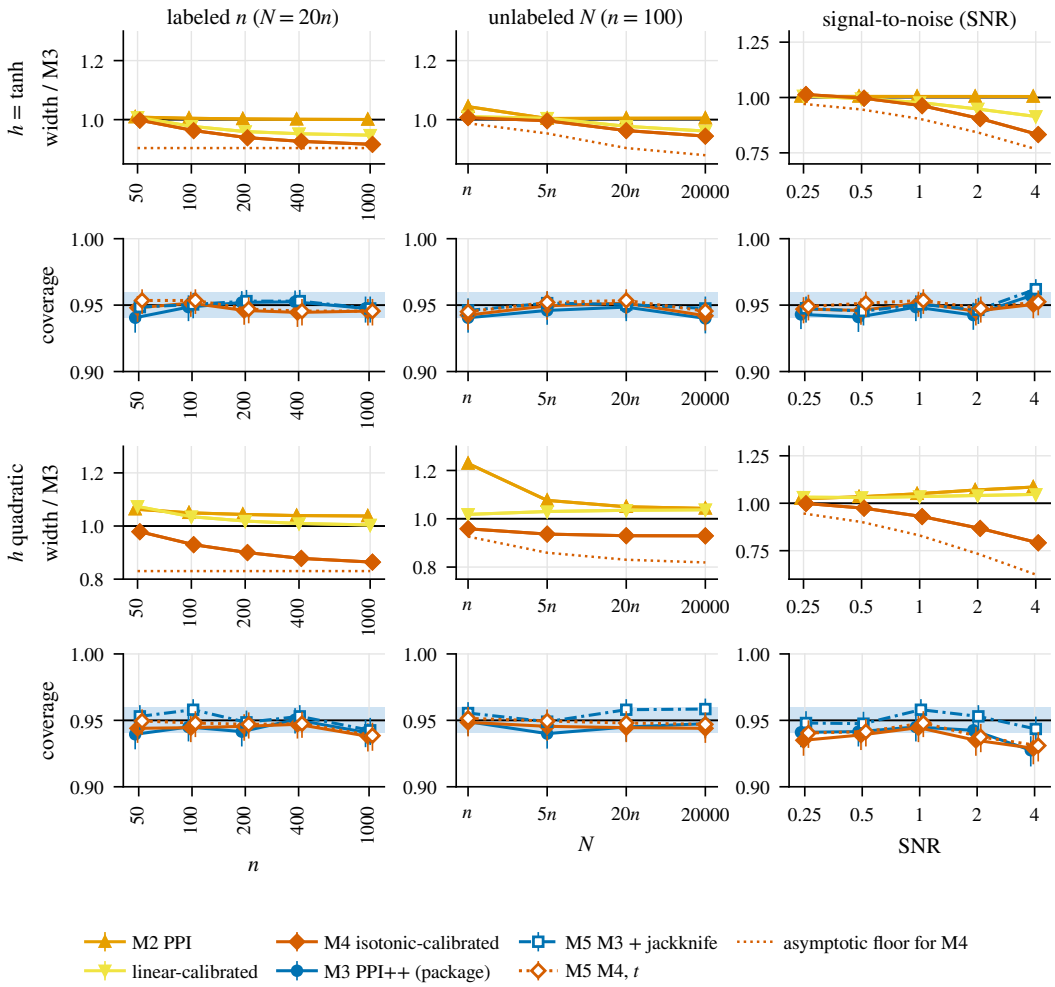


Figure 8: Synthetic populations: width ratio to M3 and coverage versus n, N and SNR ($\kappa = 0.5$).

Table 7: Heavy-tailed (Student-t_3) noise: coverage and paired width ratio of M4 to M3 (95% bootstrap interval).

h	κ	tail	M3 cov.	M4 cov.	M5 (M3) cov.	M4 / M3
quad	0.0	Student-t_3	0.950	0.952	0.958	1.061 [1.060, 1.063]
quad	0.5	Student-t_3	0.945	0.942	0.951	0.919 [0.916, 0.922]
tanh	0.0	Student-t_3	0.947	0.948	0.955	1.061 [1.060, 1.063]
tanh	0.5	Student-t_3	0.951	0.953	0.956	0.952 [0.950, 0.954]

be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.

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Table 8: Coverage at $n = 100$ and 400, $R = 2000$; bold: Wilson upper bound below 0.95. All cells: `figures/tables/tab_real_full.csv`.

Dataset	n	R	M3	M1	M2	M4	M5 (M3)	M5 (M4)
Galaxies	100	2000	**0.939**	0.945	0.950	**0.936**	0.956	**0.940**
Galaxies	400	2000	0.944	**0.940**	0.949	0.941	0.949	0.942
Forest	100	2000	**0.924**	**0.932**	0.944	**0.921**	**0.940**	**0.924**
AlphaFold	100	2000	**0.922**	**0.930**	0.945	**0.902**	0.948	**0.904**
AlphaFold	400	2000	**0.939**	0.951	0.942	**0.940**	0.945	**0.940**
Gene expression	100	2000	0.945	0.950	0.950	0.941	0.959	0.945
Gene expression	400	2000	0.947	0.948	0.948	0.949	0.951	0.951
Census income	100	2000	**0.881**	**0.909**	**0.890**	**0.889**	**0.898**	**0.891**
Census income	400	2000	**0.934**	**0.936**	**0.938**	**0.932**	0.942	**0.933**
Ballots	100	2000	**0.934**	0.942	**0.940**	**0.935**	0.946	**0.938**

Table 9: Mean width of M3 and paired width ratios to M3 ($\pm$ half the 95% bootstrap interval).

Dataset	n	M3 mean width	M1 / M3	M2 / M3	M4 / M3	M5 (M3) / M3	M5 (M4) / M3
Galaxies	100	0.118	1.457 ± 0.007	1.005 ± 0.001	1.052 ± 0.002	1.051 ± 0.002	1.065 ± 0.002
Galaxies	400	0.0592	1.450 ± 0.003	1.003 ± 0.000	1.019 ± 0.001	1.015 ± 0.001	1.022 ± 0.001
Forest	100	0.119	1.185 ± 0.004	1.010 ± 0.001	1.041 ± 0.002	1.057 ± 0.003	1.054 ± 0.002
AlphaFold	100	0.0897	1.673 ± 0.012	0.996 ± 0.001	1.073 ± 0.004	1.084 ± 0.005	1.087 ± 0.004
AlphaFold	400	0.0453	1.653 ± 0.005	0.999 ± 0.000	1.023 ± 0.001	1.026 ± 0.001	1.027 ± 0.001
Gene expression	100	0.605	2.849 ± 0.012	0.996 ± 0.001	1.042 ± 0.002	1.046 ± 0.003	1.055 ± 0.002
Gene expression	400	0.302	2.852 ± 0.006	1.000 ± 0.000	0.995 ± 0.001	1.007 ± 0.001	0.998 ± 0.001
Census income	100	22836	1.273 ± 0.005	1.004 ± 0.000	1.070 ± 0.004	1.049 ± 0.003	1.083 ± 0.004
Census income	400	11966	1.236 ± 0.002	1.001 ± 0.000	1.025 ± 0.002	1.010 ± 0.001	1.028 ± 0.002
Ballots	100	0.0539	3.592 ± 0.033	1.020 ± 0.001	1.012 ± 0.001	1.063 ± 0.006	1.024 ± 0.000

(c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

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